MSE 3114 Fall 2025 Lesson 1: Setup and First Basic Data Analysis

Overview

Duration: 2 weeks (Weeks 1-2)
Weekly Workload: 3-4 hours
Learning Focus: Anaconda environment setup, AI tool configuration and first stress-strain analysis

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Learning Objectives

By the end of this lesson, you will be able to:

- **Install Anaconda Python and required packages**
- **Configure Anaconda environment**
- **Configure AI tools** for materials science research
- **Load and validate** stress-strain data using Python
- **Perform basic analysis** with AI assistance
- **Create simple visualizations** of mechanical properties
- **Understand AI tool limitations** and best practices

📸 SCREEN CLIP INSTRUCTIONS

Throughout this lesson, you'll see this icon: 📸

When you see 📸, you need to take a screenshot or screen recording to document your progress.

How to take screen clips:

- **Windows**: Press `Windows key + Shift + S` for snipping tool
- **macOS**: Press `Cmd + Shift + 4` for area selection
- **Linux**: Press `PrtScn` or use built-in screenshot tools

Save your clips in a pdf file with descriptive captions like:

- `Cursor testing`
- `github connection`
- `python version`
- etc

Week 1: AI Tools Setup and Anaconda Python Environment Configuration

Introduction to AI-Augmented MSE 3114

Welcome to MSE 3114! This course will teach you how to use artificial intelligence tools to enhance your materials science research. We'll start by setting up the essential AI tools and then apply them to a classic materials science problem: stress-strain analysis.

Why AI-Augmented Materials Science?

Traditional materials science research involves:

- Manual data analysis and interpretation
- Time-consuming literature reviews
- Limited experimental design optimization
- Basic statistical analysis

AI augmentation provides:

- **Accelerated Analysis**: Faster data processing and interpretation
- **Enhanced Insights**: Pattern recognition beyond human capability
- **Optimized Design**: AI-assisted experimental planning
- **Automated Reporting**: Quick generation of analysis summaries

Essential AI Tools Setup

Installation Order: The order matters! Install tools in this sequence:
I. **Anaconda** (Python environment)
II. **GitHub Account** (for Copilot access)
III. **Cursor** (AI development environment)

I. Anaconda Python Environment Setup

Anaconda Installation

Step 1: Download Anaconda

1. Visit https://www.anaconda.com/
2. Click "Download" button
3. Choose your operating system:
    - **Windows**: Download the 64-bit installer (.exe file)
    - **macOS**: Download the 64-bit graphical installer (.pkg file)
    - **Linux**: Download the 64-bit installer (.sh file)

Step 2: Install Anaconda

- **Windows**: Double-click the .exe file and follow the installation wizard
    - **Important**: Check "Add Anaconda to PATH" during installation
    - **Important**: Check "Register Anaconda as default Python"
- **macOS**: Double-click the .pkg file and follow the installation wizard
- **Linux**: Open terminal and run: `bash Anaconda3-*.sh`

Step 3: Verify Anaconda Installation

Opening Command Prompt in Windows:

1. **Method 1 (Easiest)**: Press `Windows key + R`, type `cmd`, press Enter
2. **Method 2**: Press `Windows key`, type `cmd`, click "Command Prompt"
3. **Method 3**: Right-click Start button, select "Windows Terminal" or "Command Prompt"

What You'll See:

- A black window with white text will open
- You'll see something like: `C:\Users\YourName>`
- This is called the "command prompt" or "terminal"

Verify Anaconda Installation:

1. In the command prompt, type exactly: `conda --version`
2. Press Enter
3. You should see something like: `conda 23.x.x` or `conda 24.x.x`
4. 📸 *Take screenshot of your output*

If You See "conda is not recognized":

- Close the command prompt
- Restart your computer (this ensures PATH is updated)
- Try again with a new command prompt

If You Still Get Errors:

- Make sure Anaconda was installed with "Add to PATH" checked
- Try installing Anaconda again with the correct options

Step 4: Install Required Packages

Important: Use Anaconda Prompt, not regular Command Prompt!

Opening Anaconda Prompt:

1. **Method 1**: Press `Windows key`, type `Anaconda Prompt`, click "Anaconda Prompt (anaconda3)"
2. **Method 2**: Press `Windows key`, type `Anaconda`, click "Anaconda Navigator" → "Environments" → "Open Terminal"
3. **Method 3**: Start Menu → Anaconda3 → Anaconda Prompt

What You'll See:

- A blue/black window with `(base)` at the beginning
- Example: `(base) C:\Users\YourName>`
- This means you're in the base Anaconda environment

At Anaconda Prompt type:

```
# Install core scientific packages using conda
conda install pandas numpy matplotlib scipy scikit-learn

# When prompted "Proceed ([y]/n)?", type 'y' and press Enter
# Wait for installation to complete

# Install additional packages using pip
pip install opencv-python plotly streamlit polars duckdb

# Wait for all packages to install
```

Step 5: Verify Package Installation

📸 *Take screenshot of your output*

```
# Check that all packages are installed
conda list pandas numpy matplotlib scipy scikit-learn
pip list | grep -E "(opencv|plotly|streamlit|polars|duckdb)"
```

Test Installation

Use Jupyter Lab (Recommended for beginners)

1. **From Start Menu**:
    - Press `Windows key`, type `Jupyter Lab`
    - Click "Jupyter Lab (anaconda3)"

- This will open in your default web browser

What You'll See:

- A web page opens with Jupyter Lab interface
- Click the "+" button to create a new notebook
- Choose "Python 3" when prompted
- You'll see a new notebook with code cells

Test Your Environment: Copy and paste this code into a Jupyter notebook cell:

```python
# Test your environment - copy and paste this code
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import cv2
import plotly.express as px
import streamlit as st
import polars as pl
import duckdb

print("=== PACKAGE IMPORT TEST ===")
print("All packages imported successfully!")
print(f"Pandas version: {pd.__version__}")
print(f"NumPy version: {np.__version__}")
print(f"Matplotlib version: {matplotlib.__version__}")
print(f"OpenCV version: {cv2.__version__}")
print(f"Plotly version: {plotly.__version__}")
print(f"Streamlit version: {st.__version__}")
print(f"Polars version: {pl.__version__}")
print(f"DuckDB version: {duckdb.__version__}")

print("\n=== BASIC FUNCTIONALITY TEST ===")
# Test basic functionality
data = pd.DataFrame({'x': [1, 2, 3], 'y': [4, 5, 6]})
print("Pandas DataFrame created successfully:")
print(data)

# Test plotting
plt.figure(figsize=(6, 4))
plt.plot(data['x'], data['y'], 'o-')
plt.title('Test Plot')
plt.xlabel('X values')
plt.ylabel('Y values')
plt.show()

print("Environment setup complete! 🎉")
```

Running Code in Jupyter Lab:

- Click in a code cell
- Press `Shift + Enter` to run the cell
- Or click the "Run" button (play icon)
- Run cells one at a time to see output

Expected Output: You should see:

- All packages import without errors
- Version numbers displayed for each package
- A simple test plot displayed
- "Environment setup complete! 🎉" message

- 📷 *Take screenshot of your output*

If You Get Errors:

- Check that all packages were installed successfully
- Try restarting your terminal/command prompt

II GitHub Account Setup (Required for Copilot)

Step 1: Create GitHub Account

1. Visit github.com
2. Click "Sign up" and create a free account
3. Verify your email address
4. **Important**: Apply for GitHub Student Developer Pack if you're a student
   - This gives you free access to GitHub Copilot and other developer tools
   - Use your university email address for verification

III. Cursor (Primary AI Development Environment) Setup

Purpose: All-in-one AI development environment with built-in AI features
Why Cursor? Cursor is built on VS Code but includes AI features built-in, making it ideal for this course.

Setup Steps:

1. **Download Cursor**: Visit cursor.sh
2. **Install Cursor**: Run the installer for your operating system
3. **Sign in with GitHub**: This gives you access to GitHub Copilot automatically
4. **Install Python Extension**:
   - Go to Extensions (Ctrl+Shift+X)
   - Search for "Python" by Microsoft
   - Click Install
5. **Verify AI Features**:
   - Press Ctrl+K to open AI chat
   - Press Ctrl+L to open AI command palette

Cursor Advantages for This Course:

- Built-in AI chat and code generation
- Seamless GitHub Copilot integration
- AI-powered code explanation and debugging
- Better AI context understanding
- Free for students (GitHub Student Developer Pack)

2. (Optional) Additional AI Tools

ChatGPT Plus or Claude Pro (Optional Enhancement)

- **Purpose**: Additional AI research assistance beyond Cursor
- **Setup**: Only if you want extra AI help for analysis planning and interpretation
- **Cost**: $20/month subscription
- **Alternative**: Use Cursor's built-in AI features for most tasks

3. Connect Cursor to GitHub and Python

After installing Cursor, you need to connect it to your tools:

A. Connect to GitHub (Required for Copilot):

1. **Sign in to GitHub in Cursor**:

- Press `Ctrl+Shift+P` (Cmd+Shift+P on Mac)
- Type "GitHub: Sign in"
- Click "GitHub: Sign in to GitHub"
- Your default browser will open
- Authorize Cursor to access your GitHub account
- Return to Cursor when complete

2. **Verify GitHub Connection**:

   - Press `Ctrl+Shift+P`
   - Type "GitHub: Show Current User"
   - You should see your GitHub username
   - If you see "Not signed in", repeat step 1

B. Connect to Python (Required for AI Code Generation):

1. **Install Python Extension** (if not already done):

   - Press `Ctrl+Shift+X` to open Extensions
   - Search for "Python" by Microsoft
   - Click Install if not already installed

2. **Select Python Interpreter**:

   - Press `Ctrl+Shift+P`
   - Type "Python: Select Interpreter"
   - Choose "Python 3.x.x ('base': conda)" (your Anaconda Python)
   - If you don't see Anaconda, click "Enter interpreter path" and browse to:
     - **Windows**: `C:\Users\YourName\anaconda3\python.exe`
     - **macOS**: `/Users/YourName/anaconda3/bin/python`
     - **Linux**: `/home/YourName/anaconda3/bin/python`

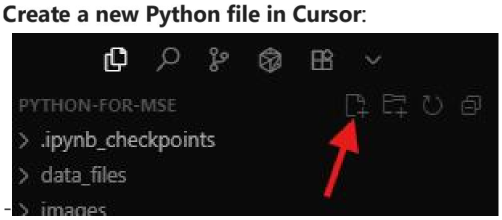
3. **Verify Python Connection**:

   - **Check bottom-right corner**: You should see "Python 3.x.x ('base': conda)"
   - **Alternative verification**: Press `Ctrl+Shift+P`, type "Python: Select Interpreter"
   - **Look for**: "Python 3.x.x ('base': conda)" in the list
   - for example: 

   - 📷 **Take screenshot showing Python interpreter selection in bottom-right corner**

C. Test the Complete Setup:

1. **Create a new Python file in Cursor**:



   - Press the New File ... button
   - The cursor will be flashing in the left panel waiting for you to type a file name.
   - Make sure you end the filename with `.py`, click Enter.
   - Press `Ctrl+S` to save it

2. **Test AI Code Generation**:

   - In the center panel (your new file), type: `# Create a pandas DataFrame with sample data`
   - Press `Ctrl+K` to open AI chat

- Ask: "Help me create a sample DataFrame with 5 rows of random data"
- The AI should generate code using your installed packages

3. **Test GitHub Copilot**:

- Type: `def calculate_stress_strain`
- Copilot should suggest function completion
- If not working, check GitHub connection in step A

First AI Interaction: Stress-Strain Analysis Planning

Let's use AI to help plan our first analysis. This demonstrates how AI can assist in research planning. In the AI chat panel on the right (in Cursor), paste the template given below.

AI Prompt Template

```
**Context**: I'm a materials science student analyzing aluminum 7075-T6 tensile test data
**Data**: CSV file with columns for load, displacement, time, and cross-sectional area
**Goal**: Perform stress-strain analysis to determine mechanical properties
**Questions**:
1. What steps should I follow for this analysis?
2. What mechanical properties can I extract?
3. What potential issues should I watch for?
4. How should I validate my results?
```

Expected AI Response

The AI should provide:

- Step-by-step analysis workflow
- List of mechanical properties to calculate
- Common data quality issues to check
- Validation methods for results

Week 1 Assignment: Environment Setup

Due: End of Week 1
Points: 10 points
Deliverables:

1. **All snipped images** from the lesson above with single sentence descriptions.
2. **AI interaction log** with your first stress-strain analysis prompt
3. **Comment** What surprised you about the AI's response?
4. **Create** a new prompt to give you "better" results?
5. **Comment** How did your improved prompt change the output?
6. **Comment** What would you do differently next time?
7. **AI interaction log** of your new edited stress-strain analysis prompt.

Submission Format: pdf file upload in Canvas

In [ ]: